

# AE Report for “Differentiable Discrete Event Simulation: Applications in Queueing Network Control”

This paper addresses gradient estimation in discrete-event simulation, focusing on job scheduling in multi-class queueing networks. The paper tackles two fundamental challenges that cause non-differentiability in sample paths:

1. The discrete nature of job assignments, where jobs must be allocated to specific servers based on queue lengths.
2. The non-differentiable `argmin` operator that determines event timing by selecting the next event from candidate arrivals and departures.

The authors propose the following solution to these two issues:

1. Allow fractional server allocations to smooth the action space.
2. Implement the “straight-through” technique from machine learning to create a differentiable approximation of the `argmin` operator.

This approach enables efficient computation of gradients for the expected cumulative cost of a job scheduling policy with respect to its parameters. A key advantage is that the gradient computation requires only a single sample path, resulting in lower variance. This is in contrast to the standard REINFORCE method—a foundation for advanced reinforcement learning algorithms—which typically requires multiple sample paths and fails to leverage the queueing system’s structural properties.

The gradient estimation method naturally integrates with stochastic gradient descent, allowing the queueing control problem to be framed as a stochastic optimization over policy parameters. Comprehensive numerical experiments demonstrate the method’s superior performance against established benchmarks.

The paper received reviews from two expert referees who acknowledge its merits while identifying areas for improvement, particularly regarding numerical experiments.

Referee 1, whose report begins with “This paper introduces a novel framework...”, requests:

- Clarification of experimental design specifications
- Extension of studies to include non-working-conserving policies and queueing control in heavy traffic regimes

Referee 2, whose report begins with “This paper introduces computational techniques...”, recommends:

- Detailed explanation of GPU parallel computing implementation
- Ablation studies to identify which components of the proposed method most significantly contribute to its performance
- Further clarification of technical details in the experimental setup

I have reviewed the paper with great interest and believe it has potential for significant impact on the simulation literature. The work’s primary strength lies in its successful adaptation of modern machine learning techniques to simulation and queueing systems, demonstrated through comprehensive numerical experiments on complex queueing control problems.

However, I agree with Referee 2’s assessment regarding the theoretical contributions, which require strengthening. As Referee 1 commented, the paper’s solutions to non-differentiable sample paths, while effective, draw heavily on existing techniques:

- The fractional server allocation method is standard.
- The “straight-through” technique for handling the `argmin` operator is borrowed directly from the machine learning literature.

The paper does introduce novel work-conserving softmax policies that effectively incorporate queueing domain knowledge into reinforcement learning frameworks. However, these policies, though innovative, are peripheral to the paper’s central focus of gradient estimation for non-differentiable sample paths.

Below are my suggestions for enhancing the theoretical contribution.

## Major Comments

1. The authors should develop a stronger theoretical framework by explicitly connecting their method to infinitesimal perturbation analysis (IPA). Both approaches share a fundamental principle: computing gradients of expected cost functions through single-sample-path analysis. As noted on page 33 (lines 30-39), the proposed method’s sample efficiency stems from its ability to provide infinitesimal counterfactuals of outcomes under alternative actions—the core intuition behind IPA (Ho and Cao, 1991, “Perturbation Analysis of Discrete Event Dynamic Systems”).

The key distinction lies in their handling of non-differentiable sample paths: the proposed method addresses this limitation while IPA does not. This suggests positioning the proposed method as a “generalized” version of IPA, similar to how Peng et al. (2018, “A New Unbiased Stochastic Derivative Estimator for Discontinuous Sample

Performances with Structural Parameters”) generalized the likelihood ratio method for non-differentiable sample paths.

To strengthen this theoretical contribution, the authors should focus on the “straight-through” technique rather than the more standard fractional server allocation approach. While the current bias-variance trade-off analysis on pages 18-19 provides good insights, it is limited to one-step transition analysis of M/M/1 systems. Extending this analysis to more complex systems would substantially enhance the theoretical framework and demonstrate the method’s broader applicability.

2. A theoretical gap exists between the paper’s methodological development and its practical application.

The proposed method is developed for a finite-horizon expected cumulative cost objective, as formalized in equations (1) and (2) on page 7. However, later discussions (page 21, line 6 and page 28, line 39 among others) indicate that the long-run average cost  $\lim_{N \rightarrow \infty} J_N(\theta)/N$  is the objective of practical interest.

While using  $J_N(\theta)$  with large  $N$  to approximate the long-run average cost is a common approach, the theoretical contribution would be substantially strengthened by analyzing the proposed method’s performance in the long-run average setting (again, with a focus on the “straight-through” technique).

3. The paper’s treatment of likelihood ratio methods requires significant revision. While the text mentions “generalized likelihood ratio” (line 15, page 12), the subsequent discussion actually describes the classic likelihood ratio method. The generalized likelihood ratio (GLR) method, introduced in Peng et al. (2018), specifically addresses gradient computation in systems with non-differentiable sample paths. This work and its subsequent developments deserve proper acknowledgment and comparative analysis, given their direct relevance to the paper’s central problem.

A valuable addition to the paper would be a comparative analysis between the proposed method and the GLR method in a setting (preferably a queueing control problem to align with this paper’s theme) where both approaches are applicable.

4. The statement on page 17 regarding the **SmoothBackprop** policy gradient estimator’s efficient computation through reverse-mode auto-differentiation using PyTorch and Jax requires elaboration. While these libraries excel at GPU-accelerated deep neural network computations, the paper doesn’t explain how the **SmoothBackprop** estimator’s structure enables their use.

The authors should:

- (i) Explain which specific mathematical structures or operations in the **SmoothBack-prop** estimator facilitate the use of reverse-mode auto-differentiation, possibly with an analogous comparison with standard neural network operations.
- (ii) Describe how the estimator’s calculations can be parallelized on GPUs.
- (iii) Clarify whether any custom operations or modifications are needed beyond standard library functions.

This technical clarification would not only strengthen the implementation details but also address Referee 2’s concerns about GPU parallel computation.

## Minor Issues

1. Page 6, paragraph 1. The statement “Generalized likelihood-ratio estimation [39]...” misattributes the generalized likelihood ratio (GLR) method. This reference (published in 1987) describes the classic likelihood ratio method, not the GLR method, which was introduced in Peng et al. (2018).
2. The discussion of IPA on page 12 lacks clarity regarding their limitations with non-differentiable sample paths. Specifically, the text should explain whether IPA’s limitation stems from its requirement for a differentiable state transition function  $f$ .
3. Page 14: What’s the difference between the fractional allocation policy, which is deterministic over continuous actions, and a randomized policy, which draws servers from a probability distribution. The work-conserving softmax policy proposed in Section 6 is indeed a randomized policy.
4. Page 26, Proposition 2: Remove duplicate “with” in the leading sentence.
5. Page 33, line 38. Typo: “... a infinitesimal...”
6. Inconsistent usage of “reinforcement learning” and its abbreviation “RL”. After the initial definition, please use the abbreviation consistently throughout.
7. Inconsistent hyphenation in “discrete event simulation” versus “discrete-event simulation”. Please standardize the hyphenation throughout the paper.
8. The paper’s exposition requires tightening, particularly in its structural organization: The core methodological contribution, presented in Section 4.2 (pages 14-19), is delayed by extensive preliminary material. This key section should be reached more efficiently to maintain reader engagement.

Section 1 (Intro) and Section 2 (Related Work) can certainly be shortened.

Section 3’s detailed description of multi-class queueing networks as discrete-event dynamic systems, while well-written, is overly elaborate. The extensive notation intro-

duced is not fully utilized in later sections, and many equations appear disconnected from the subsequent methodology and theoretical development. This section should be streamlined to emphasize only the components directly relevant to the paper’s main contributions.

9. Section 6 introduces the work-conserving softmax policy. While a novel idea, it is not central to the main theme of the paper (gradient estimation for non-differentiable sample paths). The authors may consider moving it to the appendix.
10. In both the Abstract and Introduction, the authors mention the proposed method works in “non-stationary” systems. But there’s no relevant discussion in the remaining of the paper.